

Termites

Order Isoptera

Species About 2,900

Occurrence Trees, soil, underground

These social insects feed on dead or decaying wood and live in large colonies centered on a single breeding queen. Most of the other termites in the colony either defend the nest or maintain it and care for the young. Some tropical species build spectacular mounds of earth, containing fungus gardens cultivated by the termites for their own consumption and cooled by complex ventilation systems.

Parasitic lice

Order Phthiraptera

Species 5,200

Occurrence On a variety of hosts

The wingless parasitic lice feed on the skin or blood of their hosts, which include birds, wild mammals, and humans. Their legs and strong claws are adapted for gripping feathers or hair—most species are specialized to cling to a specific type of host. They usually spend their entire lives on one animal, but may move from host to host—some spread infections in this way.

Beetles

Order Coleoptera

Species 370,000

Occurrence Terrestrial, fresh water

Almost a quarter of all known animal species are beetles, making them the most successful animals on the planet. In their adult form, most species have a single pair of wings protected by two modified forewings forming tough, often shiny, wing cases. Beetles eat a range of plant and animal matter, and some

have wood-boring larvae that can be destructive timber pests. Many have very effective chemical defenses for deterring predators.

Golden beetle
Chrysina resplendens



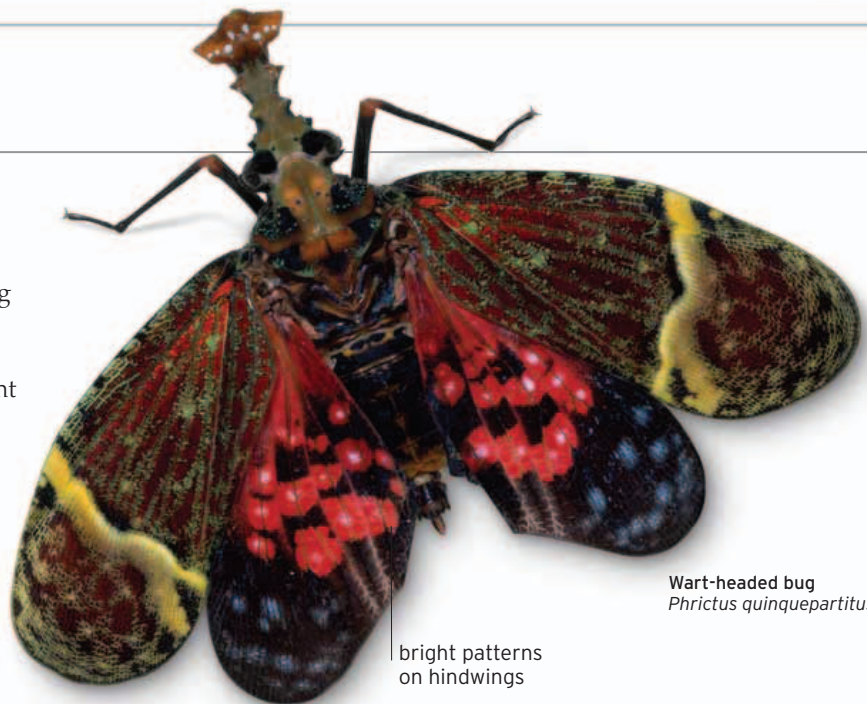
Bugs

Order Hemiptera

Species 88,000

Occurrence Terrestrial, fresh water

The true bugs are insects with piercing mouthparts for obtaining liquid food. Some, such as the aphids, cicadas, and shield bugs, are specialized for extracting plant sap. Others, such as the assassin bugs and pond skaters, attack animals—usually other insects—and inject them with digestive juices that liquefy their tissues. A few, such as the sap-sucking thorn bug, are cryptically shaped to avoid detection by birds.



bright patterns
on hindwings

Wart-headed bug
Phrictus quinquepartitus

Fleas

Order Siphonaptera

Species About 2,400

Occurrence On mammal and bird hosts, in lairs, burrows

All fleas are bloodsuckers of birds and mammals, highly specialized for life among their feathers or fur. A flea's body is flattened from side to side, enabling it to slip between hair or feather shafts, and strongly armored to resist scratching and grooming. Its long hindlegs have a unique adaptation that allow it to leap high in the air onto a new victim. In their pupal stage, fleas can lie dormant for a year or more waiting for a suitable host.

Flies

Order Diptera

Species 150,000

Occurrence Almost all habitats, namely vegetation, decaying matter, and water

Many insects are known as flies, but the true flies are distinguished by having a single pair of wings. The hindwings are reduced to a pair of club-shaped organs called halteres; these act as stabilizers and give many species phenomenal powers of flight. The group includes the well-known houseflies, blowflies, hoverflies, and mosquitoes, as well as midges, with a variety of diets and lifestyles. Some are vital pollinators of crops, while others carry deadly diseases.

Butterflies and moths

Order Lepidoptera

Species Over 165,000

Occurrence Open habitats, on vegetation

Butterflies and moths have large wings covered with small overlapping scales. The wing scales of many butterflies and day-flying moths are vividly colored in eye-catching patterns, while those of night-flying moths are cryptically colored for daytime camouflage. All species begin life as soft-bodied caterpillars that feed on leaves, growing rapidly until they turn into pupae. These develop into winged adults that sip nectar to fuel their courtship and breeding activities.

Bees, wasps, ants, and sawflies

Order Hymenoptera

Species About 198,000

Occurrence On plants, insect hosts

Notorious for their stings, but also of vital importance as pollinators, these insects are almost as numerous and diverse as the beetles. Social Hymenoptera, including ants and bees, live in highly ordered societies. All ants live in colonies controlled by a single breeding queen. Many wasps and bees—basically vegetarian wasps—do the same, including species such as hornets and

honeybees. Others are solitary, and include hunting wasps that paralyze insects and spiders with their stings and bury them as food for their young. This group also includes some parasitic species that develop inside the bodies of other insects.



German wasp
Vespa germanica